

# PHONETICS

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## 1 What is phonetics and how is it different from phonology?

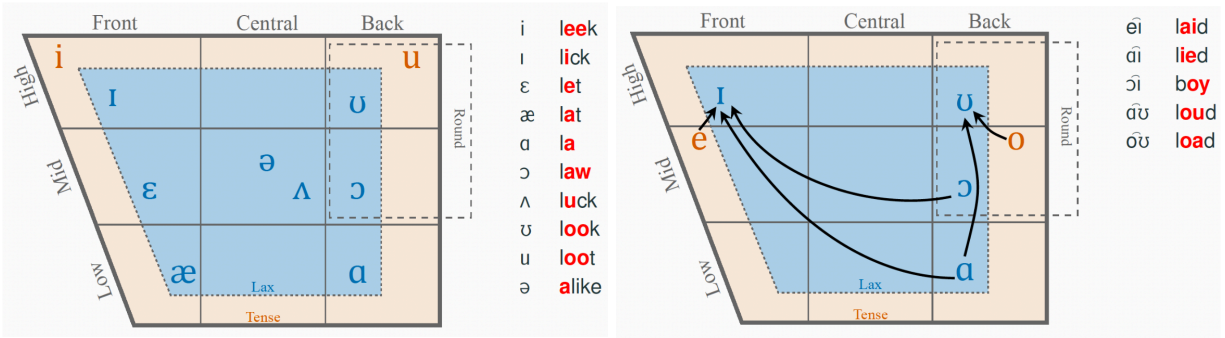
- **phonetics** is the study of the classification of speech sounds
  - it is more concerned with the description and physical properties of sounds
  - subfields of study include production, transmission, and perception
- **phonology** is the study of patterns of speech sounds
  - it is more concerned with the interaction and contrastive relationships between sounds

## 2 Basic terminology

- a **phone** is a possible speech sound; a **phoneme** is a speech sound that distinguishes meaning
  - a phoneme can have multiple phonetic variants (distinct pronunciations) called **allophones**
  - the influence of one sound on a neighboring sound is called **coarticulation**
  - phones are written with square brackets [x]; phonemes are written with slashes /x/
- there are two broad classes of speech sounds:
  - **segments** are discrete units of sound in the speech stream, e.g. vowels and consonants
  - **suprasegmentals** are features that apply to strings of segments, e.g. stress and intonation
- the two main speech segments are vowels and consonants
  - **consonants** are produced with a constriction in the vocal tract that impedes airflow
  - **vowels** allow air to flow freely through the oral cavity, or at most have only a slight narrowing

## 3 Vowels

- in the production of vowels, the airstream in the vocal tract is not turbulent
- all vowels are voiced (unless otherwise marked)
- the important properties in classifying vowels include:
  - **height** (position of tongue and jaw): high, mid, low
  - **backness** (position of tongue): front, central, back
  - **roundedness** (position of lips): rounded, unrounded
  - **tenseness** (degree of tension of tongue): tense, lax
- vowels can be divided into two categories
  - **monophthongs** are simple vowels, composed of a single configuration in the vocal tract
  - **diphthongs** are composed of two different configurations while still counting as one syllable
- in a vowel chart, when there are pairs of sounds the left is unrounded and the right is rounded
- the following are the monophthongs and diphthongs found in Standard American English:



#### 4 Consonants

- in the production of consonants, the airstream in the vocal tract is more turbulent
- the important properties in classifying consonants include:
  - **place of articulation** (position of articulators such as lips, teeth, tongue, etc.)
  - **manner of articulation** (amount of stricture and closeness of articulators)
  - **voicing** (resonance of vocal folds)
- in a consonant chart, when there are pairs of sounds the left is unvoiced and the right is voiced
- the following are the consonants found in Standard American English:

|                 |     | Place of Articulation |              |              |          |               |         |       |         |
|-----------------|-----|-----------------------|--------------|--------------|----------|---------------|---------|-------|---------|
|                 |     | Bilabial              | Labio-Dental | Inter-Dental | Alveolar | Post-Alveolar | Palatal | Velar | Glottal |
| Stop            | p b |                       |              |              | t d      |               |         | k g   | ʔ       |
| Fricative       |     |                       | f v          | θ ð          | s z      | ʃ ʒ           |         |       | h       |
| Affricate       |     |                       |              |              |          | tʃ dʒ         |         |       |         |
| Flap            |     |                       |              |              | r        |               |         |       |         |
| Nasal           |     | m                     |              |              | n        |               |         | ŋ     |         |
| Approximant     | ɹ w |                       |              |              | ɹ        |               | j       |       |         |
| Lateral Approx. |     |                       |              |              | l        |               |         |       |         |

**Voicing Quality:** Voiceless Voiced

#### 5 Suprasegmental features

- **length** is the duration that a vowel or consonant is uttered
  - some patterns in English: high vowels are shorter than low vowels; tense vowels are longer than lax vowels; voiceless consonants are longer than voiced consonants.
- **intonation** is the pattern of pitch movements across a stretch of speech
  - pitch is determined by the size of the vocal tract and the vibration of the vocal folds
  - intonation can play a role in determining meaning: a rising intonation at the end of an utterance can signal a question, a falling intonation typically signals a statement
- **tone** is the pitch at which the syllables in a word are pronounced
  - in tonal languages, tone makes a difference in a word's meaning